

Network Intrusion Detection System

Project Report

1. Introduction

In this project, I built a Network Intrusion Detection System (IDS) that classifies network traffic as either normal or malicious. The main goal was to understand how machine learning and deep learning models can be used to detect cyber attacks based on network behavior patterns.

This topic is important because cybersecurity threats are increasing, and automated detection systems help identify attacks early and improve network security.

2. Dataset Description

The dataset contains network traffic records with several features such as protocol type, source port, destination port, bytes sent, bytes received, and whether the traffic is internal or external. Each record is labeled as either normal traffic or an attack.

During preprocessing, unnecessary columns were removed, categorical features were encoded, and the dataset was split into training and testing sets to ensure proper evaluation of the models.

3. Models Used

Three machine learning models were used in this project:

- Logistic Regression
- Random Forest
- K-Nearest Neighbors (KNN)

All models were trained on the same dataset and evaluated using accuracy, precision, recall, and F1-score.

Random Forest performed the best overall due to its ability to handle complex relationships in the data. Logistic Regression gave stable results, while KNN performed the weakest among the three models.

4. Neural Network

A Neural Network was built using Dense layers with ReLU activation and a Sigmoid output layer for binary classification. Dropout layers were added to reduce overfitting and improve generalization.

The model was trained for multiple epochs and evaluated on test data. Although the neural network achieved reasonable performance, it was sensitive to feature preprocessing and required consistent input formatting.

5. Results & Comparison

Overall, Random Forest showed strong and stable performance across all metrics. The Neural Network achieved competitive results but was less stable when the input features were not perfectly aligned.

For this dataset, traditional machine learning models performed slightly better or similarly to the neural network. However, neural networks may perform better with larger datasets and more optimized feature engineering.

6. What I Learned

This project helped me understand that data preprocessing is just as important as model selection. Small differences in feature preparation can significantly affect model performance.

I also learned that more complex models are not always better for structured data. Simpler models like Random Forest can perform very well with less tuning.

Finally, I gained experience in building a full machine learning pipeline from data preprocessing to model deployment using a web interface.

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